Homework 6

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Phys 662

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Problem 1

(a) See the attached Mathematica notebook for the calculations.

Q (GeV)	α_s
4.18	0.2121
2	0.2932
1.28	0.3547

(b) The evolution of $m_f(Q)$ is given by

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}\log Q} m_f(Q) &= -\frac{2}{\pi}\alpha_s(Q) m_f(Q) \\ \implies \qquad \frac{\mathrm{d}}{\mathrm{d}Q} m_f(Q) &= -\frac{a m_f(Q)}{Q\left(1 + b\ln(Q/Q_0)\right)} \\ \implies \qquad m_f(Q) &= m_f(Q_0)\left(\frac{1}{1 + b\ln(Q/Q_0)}\right)^{a/b}. \end{aligned}$$

Substituting in the values $a = (2/\pi)\alpha_s(Q_0)$ and $b = b_0\alpha_s(Q_0)/2\pi$, we get

$$\frac{m_f(Q)}{m_f(Q_0)} = \left(\frac{\alpha_s(Q)}{\alpha_s(Q_0)}\right)^{4/b_0}$$

(c) We find a value of 1.6181 GeV for the mass of the charm quark at $Q = 2$ GeV.

flavor (GeV)	m_f (GeV)
u	0.0019
d	0.0041
s	0.0831
c	1.1078

(d) The calculated masses for the four lightest quarks at $Q = m_b$ are listed in the table above.

Problem 2

(a) See attached Mathematica notebook.

(b) Consider the operator

$$df_\pi^4 \operatorname{Tr}\left[Q_L \Sigma Q_R \Sigma^\dagger + \text{h.c.}\right],$$

where d is a dimensionless constant. This operator is $SU(3)_L \times SU(3)_R$ symmetric:

$$\begin{aligned}
\operatorname{Tr}\left[Q_L \Sigma Q_R \Sigma^\dagger\right] &\to \operatorname{Tr}\left[U_L Q_L U_L^\dagger U_L \Sigma U_R^\dagger U_R Q_R U_R^\dagger U_R \Sigma^\dagger U_L^\dagger\right] \\
&= \operatorname{Tr}\left[U_L Q_L \Sigma Q_R \Sigma^\dagger U_L^\dagger\right] \\
&= \operatorname{Tr}\left[Q_L \Sigma Q_R \Sigma^\dagger\right].
\end{aligned}$$

In the last equality, the cyclic property of the trace in used. This operator results in mass corrections

$$\delta m_{\pi^+}^2 = \delta m_{K^+}^2 = 4 d f_\pi^2$$

(c)

Problem 1 (Peskin 14.2)

In[1]:=
```
<< Notation`
```

In[2]:=
```
Symbolize[αs]; Symbolize[αs0]; Symbolize[Q0]; Symbolize[b0]; Symbolize[nf];
Symbolize[m0]; Symbolize[mf]; Symbolize[mu]; Symbolize[md];
Symbolize[ms]; Symbolize[mc]; Symbolize[mt]; Symbolize[mb];
```

In[4]:=
```
$Assumptions = {Q > 0};
```

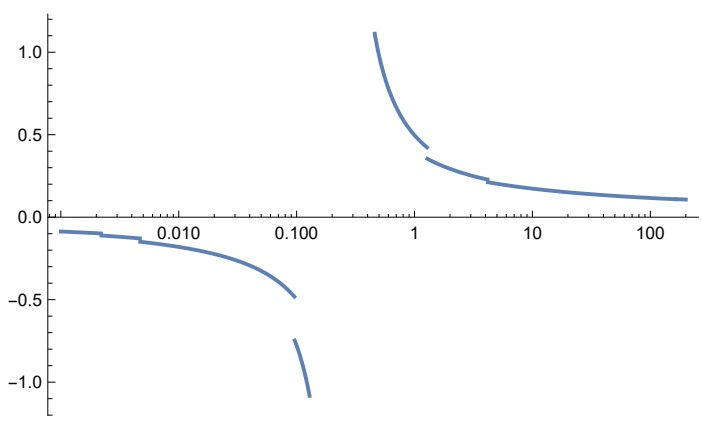
In[5]:=
```
mu = 0.0022;
md = 0.0047;
ms = 0.096;
mc = 1.28;
mb = 4.18;
mt = 164;
```

In[11]:=
$$\alpha_s = \frac{\alpha_{s0}}{1 + \left(\frac{b_0\,\alpha_{s0}}{2\,\pi}\right) \mathrm{Log}[Q / Q_0]};$$

In[12]:=
```
Q0 = 91;
αs0 = 0.118;
nf = Piecewise[{{0, Q < mu}, {1, mu ≤ Q < md}, {2, md ≤ Q < ms},
    {3, ms ≤ Q < mc}, {4, mc ≤ Q < mb}, {5, mb ≤ Q < mt}, {6, mt ≤ Q}}];
b0 = 11 - 2/3 nf;
```

In[16]:=
```
LogLinearPlot[αs, {Q, 0.001, 200}]
```

Out[16]=



a)

In[17]:=
```
Q = m_b;
α_s
```
Out[18]=
```
0.212056
```

In[19]:=
```
Q = 2;
α_s
```
Out[20]=
```
0.293154
```

In[21]:=
```
Q = m_c;
α_s
```
Out[22]=
```
0.354702
```

b)

In[23]:=
```
ClearAll[Q, Q_0, n_f, α_s0, b_0];
```

In[24]:=
```
m_f = m[Q] /. DSolve[{m'[Q] == -2/π α_s m[Q]/Q, m[Q_0] == m_0}, m[Q], Q][[1]] // Simplify
```
Out[24]=

$$m_0 \left(1 + \frac{b_0\, \alpha_{s0}\, \mathrm{Log}\left[\frac{Q}{Q_0}\right]}{2\,\pi}\right)^{-4/b_0}$$

In[25]:=
```
n_f = Piecewise[{{0, Q < m_u}, {1, m_u ≤ Q < m_d}, {2, m_d ≤ Q < m_s},
    {3, m_s ≤ Q < m_c}, {4, m_c ≤ Q < m_b}, {5, m_b ≤ Q < m_t}, {6, m_t ≤ Q}}];
b_0 = 11 - 2/3 n_f;
```

c)

In[27]:=
```
Q = 2;
Q_0 = m_c;
α_s0 = 0.35470157484375525;
m_0 = m_c;
m_f
```
Out[31]=
```
1.1681
```

d)

In[32]:=
```
Q = m_b;
Q_0 = 2;
α_s0 = 0.35470157484375525;
m_0 = m_u;
m_f
m_0 = m_d;
m_f
m_0 = m_s;
m_f
m_0 = m_c;
m_f
```

Out[36]=
```
0.00190405
```

Out[38]=
```
0.00406774
```

Out[40]=
```
0.0830858
```

Out[42]=
```
1.10781
```

Problem 2

In[1]:=
```
<< Notation`
```

In[2]:=
```
Symbolize[π⁰]; Symbolize[π⁺]; Symbolize[π⁻];
Symbolize[K⁰]; Symbolize[K⁺]; Symbolize[K⁻]; Symbolize[K̄⁰];
Symbolize[η]; Symbolize[f_π]; Symbolize[Σ⁺];
Symbolize[m_u]; Symbolize[m_d]; Symbolize[m_s];
Symbolize[Q_L]; Symbolize[Q_R];   Symbolize[L_2];
```

In[7]:=
```
$Assumptions = {m_u ∈ ℝ, m_d ∈ ℝ, m_s ∈ ℝ, f_π ∈ ℝ};
```

In[8]:=

$$\Pi = \left\{\left\{\pi^0 + \frac{1}{\sqrt{3}}\eta,\ \sqrt{2}\,\pi^+,\ \sqrt{2}\,K^+\right\},\ \left\{\sqrt{2}\,\pi^-,\ -\pi^0 + \frac{1}{\sqrt{3}}\eta,\ \sqrt{2}\,K^0\right\},\ \left\{\sqrt{2}\,K^-,\ \sqrt{2}\,\bar{K}^0,\ \frac{-2}{\sqrt{3}}\eta\right\}\right\};$$

```
M = DiagonalMatrix[{m_u, m_d, m_s}];
MatrixForm /@ {Π, M} // Row
```

Out[10]=

$$\begin{pmatrix} \frac{\eta}{\sqrt{3}} + \pi^0 & \sqrt{2}\,\pi^+ & \sqrt{2}\,K^+ \\ \sqrt{2}\,\pi^- & \frac{\eta}{\sqrt{3}} - \pi^0 & \sqrt{2}\,K^0 \\ \sqrt{2}\,K^- & \sqrt{2}\,\bar{K}^0 & -\frac{2\eta}{\sqrt{3}} \end{pmatrix} \begin{pmatrix} m_u & 0 & 0 \\ 0 & m_d & 0 \\ 0 & 0 & m_s \end{pmatrix}$$

In[11]:=

$$n = 2;$$

$$\Sigma = \sum_{i=0}^{n} \frac{1}{i!}\left(\frac{i}{f_\pi}\right)^i \text{MatrixPower}[\Pi, i];$$

$$\Sigma^+ = \sum_{i=0}^{n} \frac{1}{i!}\left(\frac{-i}{f_\pi}\right)^i \text{MatrixPower}[\Pi, i];$$

```
MatrixForm /@ {Σ, Σ⁺} // Row
```

Out[14]=

$$\begin{pmatrix} 1 + \frac{i\left(\frac{\eta}{\sqrt{3}}+\pi^0\right)}{f_\pi} - \frac{2K^-K^+ + \left(\frac{\eta}{\sqrt{3}}+\pi^0\right)^2 + 2\pi^-\pi^+}{2f_\pi^2} & \frac{i\sqrt{2}\,\pi^+}{f_\pi} - \frac{2\bar{K}^0K^+ + \sqrt{2}\left(\frac{\eta}{\sqrt{3}}-\pi^0\right)\pi^+ + \sqrt{2}\left(\frac{\eta}{\sqrt{3}}+\pi^0\right)\pi^+}{2f_\pi^2} & \frac{i\sqrt{2}\,K^+}{f_\pi} - \frac{-2\sqrt{\frac{2}{3}}\,K^+\eta + \sqrt{}}{} \\ \frac{i\sqrt{2}\,\pi^-}{f_\pi} - \frac{2K^0K^- + \sqrt{2}\left(\frac{\eta}{\sqrt{3}}-\pi^0\right)\pi^- + \sqrt{2}\left(\frac{\eta}{\sqrt{3}}+\pi^0\right)\pi^-}{2f_\pi^2} & 1 + \frac{i\left(\frac{\eta}{\sqrt{3}}-\pi^0\right)}{f_\pi} - \frac{2\bar{K}^0K^0 + \left(\frac{\eta}{\sqrt{3}}-\pi^0\right)^2 + 2\pi^-\pi^+}{2f_\pi^2} & \frac{i\sqrt{2}\,K^0}{f_\pi} - \frac{-2\sqrt{\frac{2}{3}}\,K^0\eta + \sqrt{}}{} \\ \frac{i\sqrt{2}\,K^-}{f_\pi} - \frac{-2\sqrt{\frac{2}{3}}\,K^-\eta + \sqrt{2}\,K^-\left(\frac{\eta}{\sqrt{3}}+\pi^0\right) + 2\bar{K}^0\pi^-}{2f_\pi^2} & \frac{i\sqrt{2}\,\bar{K}^0}{f_\pi} - \frac{-2\sqrt{\frac{2}{3}}\,\bar{K}^0\eta + \sqrt{2}\,\bar{K}^0\left(\frac{\eta}{\sqrt{3}}-\pi^0\right) + 2K^-\pi^+}{2f_\pi^2} & 1 - \frac{2i\eta}{\sqrt{3}\,f_\pi} - \frac{2\bar{K}^0}{} \end{pmatrix}$$

a)

In[15]:=
```
L = c f_π^3 Tr[M.Σ + M†.Σ⁺] // FullSimplify
```

Out[15]=

$$\frac{1}{3} c f_\pi \left(-6 \bar{K}^0 K^0 (m_d + m_s) - 6 K^- K^+ (m_s + m_u) + 6 f_\pi^2 (m_d + m_s + m_u) - (m_d + 4 m_s + m_u) \eta^2 + 2 \sqrt{3} (m_d - m_u) \eta \pi^0 - 3 (m_d + m_u) \left(\pi^0\right)^2 - 6 (m_d + m_u) \pi^- \pi^+\right)$$

In[16]:=
```
-Coefficient[L, K⁺ K⁻]
```

Out[16]=

$$2 c f_\pi (m_s + m_u)$$

In[17]:=
```
-Coefficient[L, K⁰ K̄⁰]
```

Out[17]=

$$2 c f_\pi (m_d + m_s)$$

In[18]:=
```
-2 Coefficient[L, η^2] // Simplify
```

Out[18]=

$$\frac{2}{3} c f_\pi (m_d + 4 m_s + m_u)$$

b)

In[19]:=
```
Q_L = DiagonalMatrix[{2/3, -1/3, -1/3}];
Q_R = Q_L;
```

In[21]:=
```
L_2 = d f_π^3 Tr[Q_L.Σ.Q_R.Σ⁺ + Σ.Q_R.Σ⁺.Q_L] // FullSimplify
```

Out[21]=

$$\frac{1}{54 f_\pi} d \left(7 \eta^4 + 4 \sqrt{3} \eta^3 \pi^0 + 6 \eta^2 \left(4 \bar{K}^0 K^0 + 4 K^- K^+ + 5 \left(\pi^0\right)^2 - 2 \pi^- \pi^+\right) + 12 \sqrt{3} \eta \left(\left(\pi^0\right)^3 - \sqrt{2} \left(\bar{K}^0 K^+ \pi^- + K^0 K^- \pi^+\right) + 2 \pi^0 \left(2 K^- K^+ + \pi^- \pi^+\right)\right) + 3 \left(24 f_\pi^4 + 8 \left(\bar{K}^0\right)^2 \left(K^0\right)^2 + 5 \left(\pi^0\right)^4 - 12 \sqrt{2} \pi^0 \left(\bar{K}^0 K^+ \pi^- + K^0 K^- \pi^+\right) - 72 f_\pi^2 \left(K^- K^+ + \pi^- \pi^+\right) + 4 \left(\pi^0\right)^2 \left(2 \bar{K}^0 K^0 + 2 K^- K^+ + 5 \pi^- \pi^+\right) + 4 \left(K^- K^+ + \pi^- \pi^+\right) \left(-2 \bar{K}^0 K^0 + 5 K^- K^+ + 5 \pi^- \pi^+\right)\right)\right)$$

In[22]:=
```
-Coefficient[L_2, K⁺ K⁻] /. {π⁰ → 0, π⁺ → 0, π⁻ → 0, K⁰ → 0, K̄⁰ → 0, η → 0}
```

Out[22]=

$$4 d f_\pi$$

In[23]:=
```
-Coefficient[L_2, π⁺ π⁻] /. {π⁰ → 0, K⁺ → 0, K⁻ → 0, K⁰ → 0, K̄⁰ → 0, η → 0}
```

Out[23]=

$$4 d f_\pi$$

In[24]:=
```
-Coefficient[L2, K0 K̄0] /. {π0 → 0, π+ → 0, π- → 0, K+ → 0, K- → 0, η → 0}
```
Out[24]=
```
0
```

In[25]:=
```
-Coefficient[L2, π0 π0] /. {π+ → 0, π- → 0, K+ → 0, K- → 0, K0 → 0, K̄0 → 0, η → 0}
```
Out[25]=
```
0
```

In[26]:=
```
-2 Coefficient[L2, η^2] /. {π0 → 0, π+ → 0, π- → 0, K+ → 0, K- → 0, K0 → 0, K̄0 → 0}
```
Out[26]=
```
0
```